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Batch : S23

Subject : Fundamentals of Telecommunication Systems

Experiment Number : 4

Experiment Title : Demonstration of Modem Hardware and Signal Flow

Date of Performance : 09/02/2026

Date of Submission : 30/03/2026

Experiment No. 04: Demonstration of Modem Hardware and Signal Flow

Aim

To identify the hardware interfaces of a Modem and demonstrate the flow between a PC and an ISP Server using Cisco Packet Tracer.

Theory

What is a Modem?

A Modem (Modulator–Demodulator) is a networking device that converts digital signals generated by a computer into analog signals for transmission over a communication medium such as a coaxial cable. It also converts incoming analog signals back into digital form so that the computer can process the data.

In this experiment, the Cable Modem-PT acts as an interface between the PC and the ISP cloud, enabling communication between the digital Ethernet network and the analog WAN infrastructure.

Explain the working principle of a modem.

The working principle of a modem is based on **modulation and demodulation**.

Modulation (Sending Process):

- The PC generates digital data (0s and 1s).
- The modem converts this digital data into analog signals.
- The analog signal is transmitted through the coaxial cable to the ISP cloud.

Demodulation (Receiving Process):

- The modem receives analog signals from the ISP.
- It converts these analog signals back into digital form.
- The PC processes the received digital data.

In this experiment, when a ping command is executed:

- The signal travels digitally from PC to Modem via Ethernet.

  The Modem converts it for transmission over Coaxial cable.

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- The ISP cloud maps the connection and forwards it to the Server.
- The reply follows the same path in reverse.

Thus, the modem enables communication between digital end devices and analog WAN infrastructure.

What is IP address?

An **IP Address (Internet Protocol Address)** is a unique logical address assigned to each device on a network.

It is used to:

- Identify a device on a network.
- Enable communication between devices.
- Ensure correct delivery of data packets.

In this experiment:

- PC IP Address: 192.168.1.1
- Server IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0

Both devices belong to the same network, allowing direct communication.

What is ISP?

ISP (Internet Service Provider) is an organization that provides Internet connectivity and related services to users.

In this experiment, the ISP is represented by the Cloud-PT device, which simulates the service provider's infrastructure. It maps the coaxial interface to the Ethernet interface and allows communication between the PC and the Server.

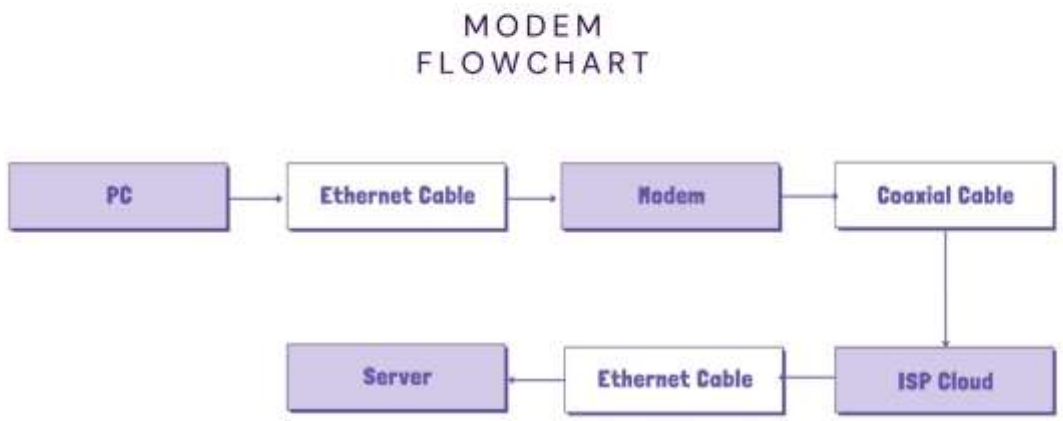
Examples of real-world ISPs include:

- Jio
- Airtel

- BSNL

An ISP assigns IP addresses and provides the infrastructure required for Internet communication.

Block Diagram



Hardware Components Required

- **PC-PT:** The user's computer (Digital Source).
- **Cable Modem-PT:** The signal converter (Modulator-Demodulator).
- **Cloud-PT (ISP):** The Service Provider infrastructure.
- **Server-PT:** The remote destination (e.g., a Web Server).

Procedure

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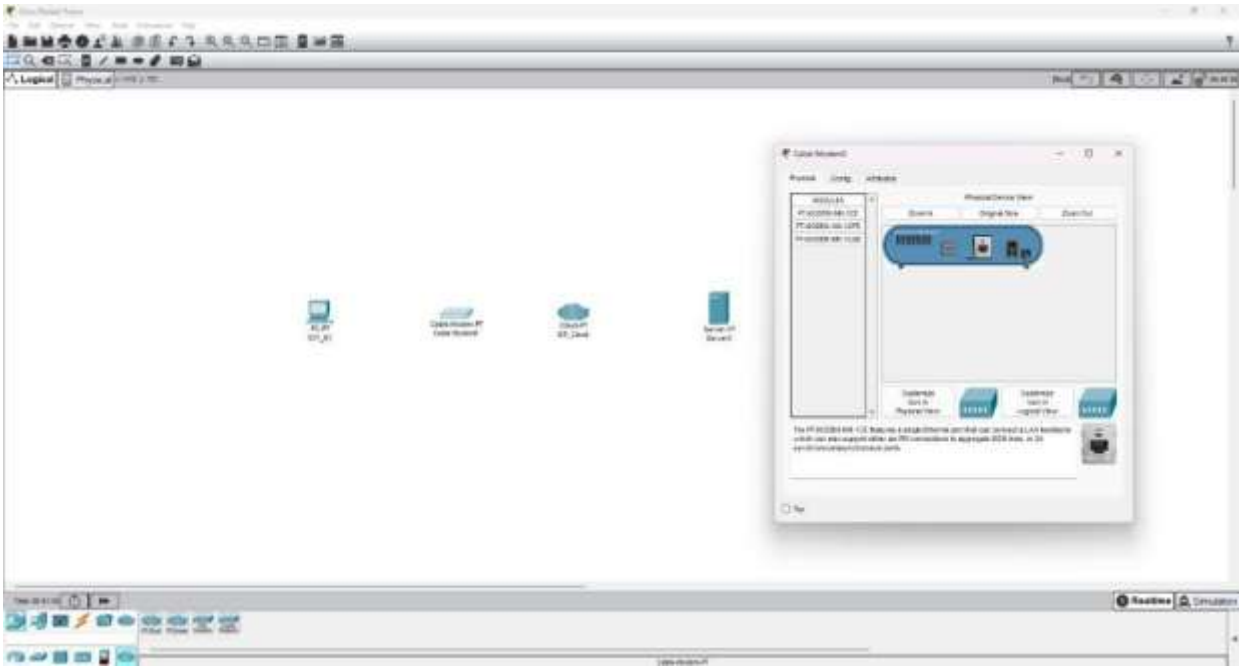
Step 1: Physical Topology Design

In this stage, we select the hardware from the device box and place them on the logical workspace.

Add End Devices: Go to the bottom-left "End Devices" category. Drag and drop one PC (PCPT) and one Server (Server-PT) onto the workspace. Rename the PC by your batch and roll number (e.g. S11_01).

- **Add the ISP Infrastructure:** From the "WAN Emulation" category, drag and drop the CloudPT between the PC and the Server. Rename this as "ISP_Cloud".
- **Add the Modem:** Go to "Network Devices" -> "WAN Emulation". Drag and drop the Cable Modem-PT between the PC and the Cloud.
- **Hardware Identification:** Click on the Modem and observe the physical rear view. Note the difference between the small RJ-45 Ethernet port and the threaded Coaxial port. • Rear View



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Step 2: Physical Cabling (Media Selection)

Selecting the correct transmission media is crucial for simulating real-world telecommunication links.

- **PC to Modem:** Select the Copper Straight-Through cable (solid black line). Connect the FastEthernet 0 port of the PC to Port 1 (ethernet port) of the Cable Modem.

  Modem to ISP: Select the Coaxial cable (solid blue line). Connect Port 0 (co-axial port) of the Cable Modem to the Coaxial 7 port of the ISP_cloud.

- **ISP to Server:** Select the Copper Straight-Through cable. Connect the Ethernet 6 port of the ISP_Cloud to the FastEthernet 0 port of the Server.

Step 3: ISP Cloud Logic Configuration

The Cloud must be told how to "route" the signal from the analog coaxial side to the digital server side.

- Click on the ISP_Cloud and go to the Config tab.
- In the left-hand menu, look under the INTERFACE heading and click on Ethernet6. Ensure "Cable" is selected under the Provider Network.
- In the left-hand menu, look under the CONNECTIONS heading and click on Cable.
- In the drop-down menus, select Coaxial 7 for the first field and Ethernet 6 for the second field.
- Click the Add button. You should now see Coaxial7 <==> Ethernet6 appearing in the mapping table.

Step 4: IP Addressing (Logical Setup)

Both the source (PC) and destination (Server) must belong to the same logical network to communicate without a router.

- **Configure PC:** Click PC -> Desktop -> IP Configuration. Set the Static IP address to 192.168.1.1 and Subnet Mask to 255.255.255.0.
- **Configure Server:** Click Server -> Desktop -> IP Configuration. Set the IP address to 192.168.1.2 and Subnet Mask to 255.255.255.0.

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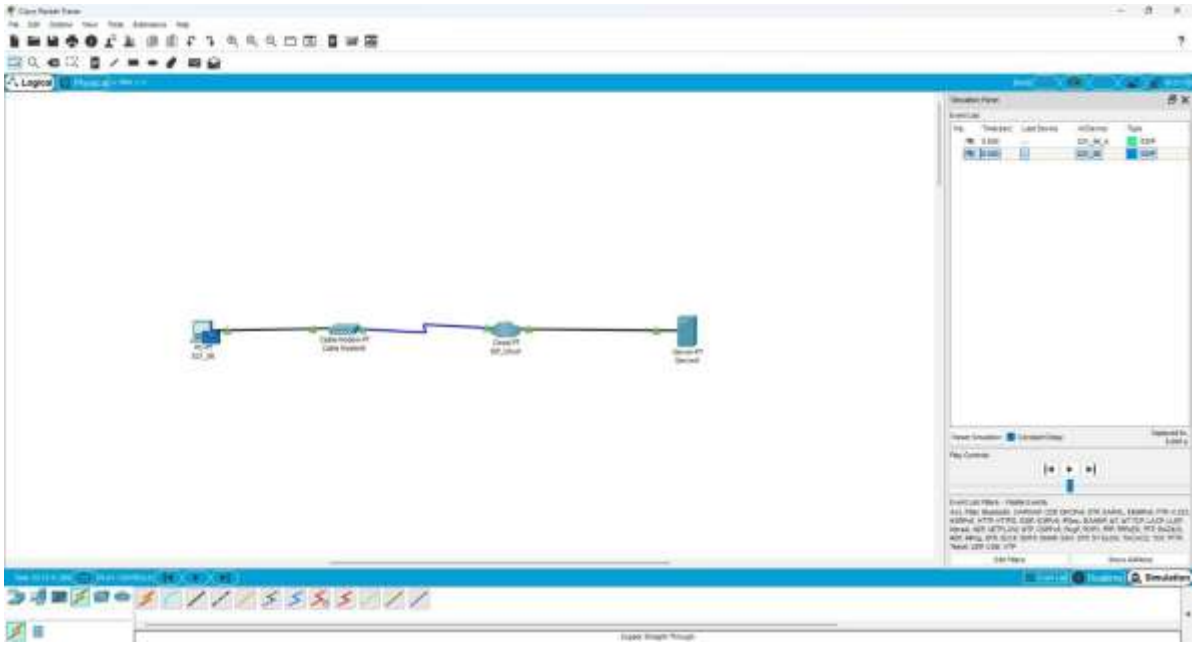
Step 5: Testing and Packet Observation

- **Real-time Testing:**
  - Open the Command Prompt on the PC. Type ping 192.168.1.2. If you see "Reply from...", the modem is correctly modulating and demodulating the signal.
- **Simulation Mode:** Click the Simulation tab (bottom-right).
  - Click Edit Filters and select only ICMP. ○ Click the Add Simple PDU (Envelope icon), click on the PC, then click on the Server.

- Click the Play/Forward button to observe the packet travel from the digital Ethernet link to the analog Coaxial link.



Observations

- **Medium Change:** It is observed that the data travels as a digital signal until it reaches the Modem.
- **Signal Conversion:** Upon hitting the Modem, the data is converted to travel over the **Coaxial medium** (simulating analog transmission).
- **Successful Loop:** The Server successfully receives the modulated signal, ISP cloud demaps the return packet back to the specific Coaxial line, allowing the PC to receive a successful reply.

Conclusion

The experiment successfully demonstrates that a Modem is an essential interface for internet connectivity. It proves that digital end-devices cannot communicate directly over analog WAN infrastructure without the modulation/demodulation hardware.